MILITARY SPECIFICATION

RESISTORS, VARIABLE, NONWIRE-WOUND, STYLE RVC5

This specification is approved for use by all Departments and Agencies of the Department of Defense.

INACTIVE FOR DESIGN AFTER DATE OF THIS SPECIFICATION (SEE 6.3).

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RVC5 resistors.

2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-R-23285 - Resistors, Variable, Nonwire-Wound, General Specification for.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-23285 only, as specified herein.

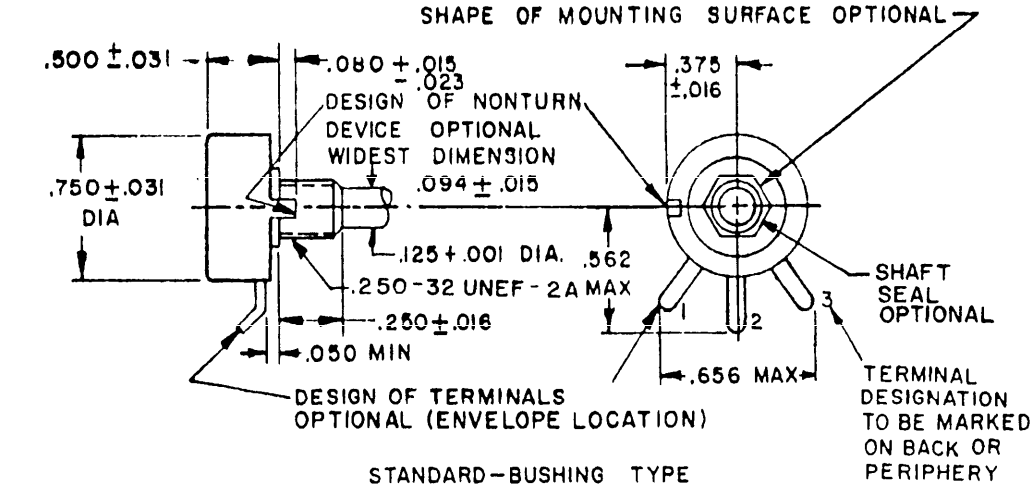
3.2 Design and construction. Resistors shall be of the design, construction, and physical dimensions shown on figure.

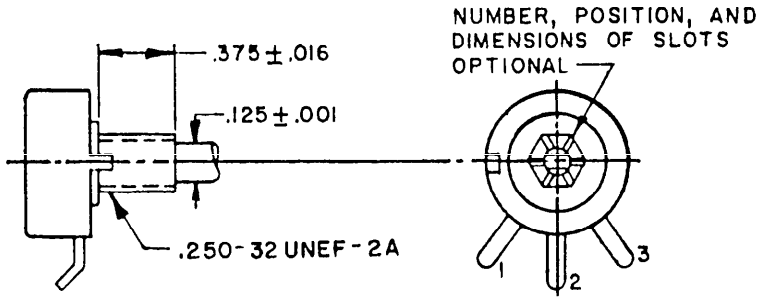
3.2.1 Mounting and locking nuts. The mounting nut shall be 5/64-inch thick and shall measure 5/16 inch across the hexagonal flats. The locking nut shall be 5/32-inch thick and shall measure 5/16 inch across the hexagonal flats. Thread size shall be 1/4-32NEF-2B.

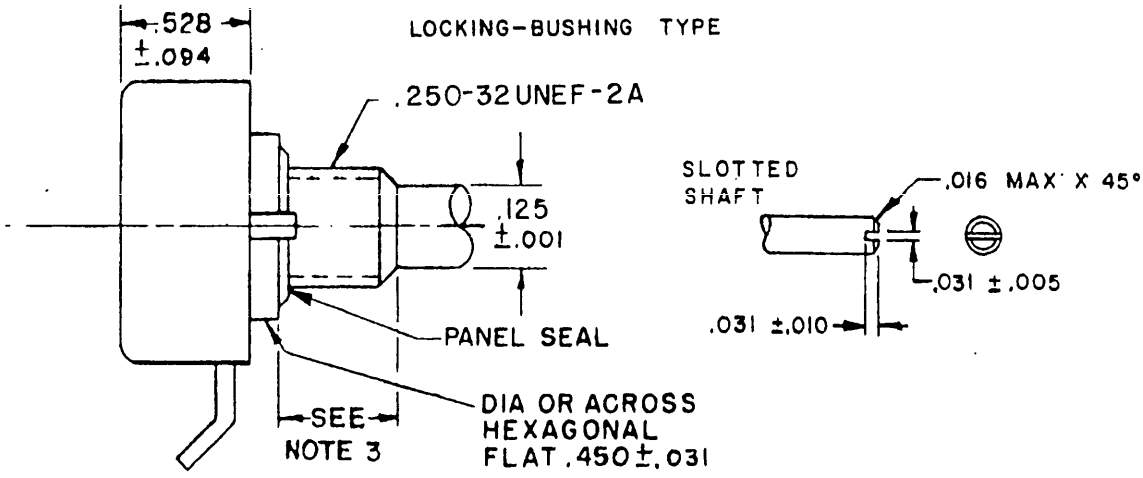
3.2.2 Internal-tooth lockwasher. Internal-tooth lockwasher of suitable size shall be supplied.

3.2.3 Retainer rings. If retainer rings are used, they shall be not thicker than 0.032 inch.

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SHAFT-AND PANEL-SEALED LOCKING AND STANDARD BUSHING TYPE

STYLE RVC5 RESISTORS

NOTES:
1. Dimensions are in inches.
2. Thread length is .250 ±.016 for standard bushing and .375 ±.016 for locking bushing.

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TABLE I. Length of operating shaft.

Symbol	Nominal length of shaft measured from mounting surface of resistors 1/	
	Bushings N and S Inches ±1/32	Bushings L and T Inches ±1/32
B	---	1/2
D	7/8	7/8
L	3/8	---

1/ For resistors with panel seals. The mounting surface shall be considered the metal face of the panel seal when the seal is firmly seated against the resistor body.

3.3 Total mechanical rotation. The total mechanical rotation shall be within the limits of 275° to 285°.

3.4 Nominal total resistance and rated continuous working voltages. The nominal total resistance and rated continuous working voltage (RCWV) shall be as specified in table II.

TABLE II. Nominal total resistance and rated continuous working voltage.

Nominal total resistance	RCWV 1/ (volts)		Nominal total resistance	RCWV 1/ (volts)		Nominal total resistance	RCWV 1/ (volts)	
	Taper-A	Taper-C		Taper-A	Taper-C		Taper-A	Taper-C
Ohms			Ohms			Megohms		
50	7	---	5,000	71	50	.10	325	200
100	10	---	7,500	89	62	.15	350	200
150	12	---	10,000	100	71	.25	350	200
250	16	---	15,000	125	89	.35	350	200
350	19	---	25,000	160	112	.50	350	200
500	23	---	35,000	190	135	.75	350	200
750	27	---	50,000	250	158	1.0	350	200
1,000	31	22	75,000	275	200	1.5	350	---
1,500	40	26				2.0	350	---
2,500	51	35				2.5	350	---
3,500	60	42				3.5	350	---
						5.0	350	---

1/ Rated continuous working voltage at 125°C.

3.5 Maximum weight. The maximum weight shall be 30 grams, including hardware.

3.6 Power rating. The power rating shall be 1 watt for taper-A resistors and 1/2 watt for taper-C resistors.

3.7 Torque.

3.7.1 Operating. The torque required to effect rotation shall be not less than 0.50 ounce-inch nor more than 6 ounce-inches.

3.7.2 Stop. The torque applied through the operating shaft to the stops shall be 5 pound-inches.

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3.7.3 Locking bushings. For the locking-bushing type, the locking nut shall be tightened with a torque of 10 pound-inches. After the locking nut is tightened, the contact arm shall not move when a torque of 20 ounce-inches is applied to the shaft.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Qualification inspection. Not applicable to this specification.

* 4.2 General. The sampling, acceptance inspection, and test procedures specified in tables VII and VIII of MIL-R-23285 are applicable as sampling, inspection, and test procedures of this detail specification.

4.3 Dielectric withstanding voltage. In the dielectric withstanding voltage test, the applied potential at reduced barometric pressure shall be 350 volts root mean square (rms).

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-R-23285.

6. NOTES

6.1 Notes. The notes specified in MIL-R-23285 are applicable to this specification.

6.2 This specification style RVC5 supersedes style RVC5 of MIL-R-23285/2A(NAVY), dated 22 November 1965, and style RVC5 of MIL-R-23285/2B, dated 13 September 1965.

* 6.3 Resistors covered by this specification are for replacement only and not for new design.

6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - MI, MU
Navy - AS, OS
Air Force - 17, 85
DSA - ES

User activities:
Army - AT, AV, ME
Navy - MC
Air Force - 19

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5905-0884-1)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-R-23285/2D RESISTORS, VARIABLE, NONWIRE-WOUND, STYLE RVC5

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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